Video Transcript

Video 10: Creating a Loan Review Workflow (07:18)

Let's try to do a workflow that has more components. Let's try to mockup something like a loan review type of workflow. With that in mind, let's go ahead and move to the editor. We'll start off by creating a new file. We'll 'Select the language' of Python. We'll go ahead and save it. In this case into the sample directory that we have been working on. In this case, I'll call it loan_review. We'll start out as usual by doing the imports. Then the operators. Notice, there that we have an additional one. This case, in addition to the BashOperator, the PythonOperator, we have a BranchPythonOperator that we will use to make a selection between a couple of options. We will see that subsequently here in the code.

Then we're going to add some packages for some functions that we're going to be using. We'll then do our dag instance there. As you can see, we'll call it loan review. It'll start a day ago and it'll run every one day. Then we're going to start off with our tasks. And as you can see here, we have three loans that we will be comparing, and we will be calling a Python function that will predict whether this is a good loan or a bad loan. And so, we have loan A, loan B, and loan C. After that we're going to have our branch that is going to select from those three loans by calling a function that we will write as well afterwards called best loan. And depending on that, it'll write either good loan or bad loan, in this case, just good or bad. So, let's go to those functions, the very first one. And actually, before we do that, let me go ahead and show you here the sequence. We're going to run loan A, B, and C in parallel, since those are simply going to give us a score and we'll see that in a moment how that is done.

Then we're going to have that branch, a task that is going to choose the best loan, and then we're going to write out good or bad depending on what we get there. So, the next step is that we're going to write our best loan. And actually, we need to also look at our prediction. So, our prediction, as you can see there, since we're doing a mock here, is simply a function that returns a random integer between one and ten. And we're going to take anything greater than eight, as you can see in the function above to be a good loan. Now from the very first three tasks that run, so loan A, loan B, and loan C, these here, we're going to get a value. The value is going to be

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that prediction, the result of calling the prediction function, the function that will return an integer between 1 and 10.

Now to be able to get access to the output of each of those tasks, we're going to use this parameter called ti, and we're going to pull from it the result of those tasks IDs. So, loan A, loan B, loan C. Again, this is what we called loan those blocks, those tasks. So, loan A, loan B, loan C. And then, we're going to create an array from that right of results, the predictions. And then from it we're going to pick the one with the best value or the highest value. Now here in this conditional statement, we're simply going to check and see if that value is greater than eight, and if it is, we're going to return good, otherwise we're going to return bad. So, that completes the logic that we're capturing here. Again, it's a mock of being able to select amongst a number of values, in this case three, those three loans, and then we're going to branch here choosing the best one.

We just saw the function that performs that, which is the function called best_loan. And then we're simply going to select good or bad depending on what the output is. So, let's go ahead and save this. And now, we're going to drag and drop that into our dags folder. If we go into it, we can see that it is there now. And now we're going to go ahead and wait for the Airflow platform to pick up that DAG. And as you can see here, I have waited for my platform to load the DAG. I'm going to go into loan review. I'm going to go ahead and activate it. Actually, before I do, let me go ahead and show you here. You can see, how loan A and the options that it has. And if we look at the graph view, which is more intuitive in this case, you can see that loan A, B, and C is running in parallel. And then we have that branching and then the output of good or bad.

So, let's go ahead and activate this, and let's go ahead and run it or trigger it by hand. As you can see there, we immediately have a result, and we have two. Now let's go ahead and take a look at the graph view. And take a look at the output of each of these. We can go ahead and look at the log. This return four. So, let's go ahead and look at B. This was 10, so this would be considered a good loan. Let's go ahead and look at C. And in this case, we have 6. So, that would also be a bad loan such as the first one. Now we're going to go ahead and take a look here at the selection. And you can see here that one of those choices was good, the one that we had a ton. And then, you can see here that we have a good option and log capturing that result echoing good, you can see there.

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And if we look at the other one, you'll see here that it did not run because it didn't meet, it didn't select that branch. So, as you can see there, that's a pretty representative problem of the type of wiring you might want to do. You can see here, how that is a very easy way to hook onto the functionality of what this platform provides you. As mentioned before, you would have to build this anyway yourself as you grew a script that was homegrown onto something that was relied upon by many people or perhaps many departments. And so, having this in place to be able to compose those workflows is something that is pretty useful. So now, it's your turn go ahead and make sure that you can run this example explorer on your own as well a few variations and get familiar with this platform as it is something that can be quite useful.

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